



SHEAR, PRESSURE, AND VIBRATION COMFORT SYSTEM

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Project Goal

Develop a device and testing method for automotive seating that provides quantitative results to indicate occupant long-term comfort for at least 45 minutes, without the use of a live test subject. In order to fulfill this, the following needed to be completed:

- Define parameters affecting long term seat comfort.
- Develop testing method of parameters.
- Physically measure parameters.
- Validate measurements with human trials.

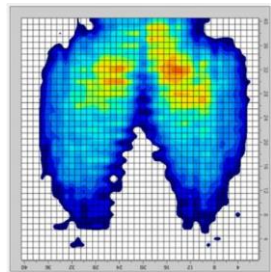
Background

Comfort requirements:

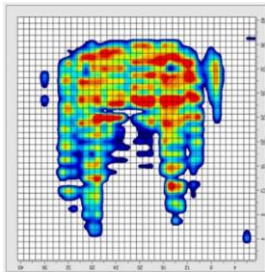
- Physical
- Physiological
- Psychological perception

Current seat testing:

- Creating force maps
- Evaluating seat geometry
- Quantifying seat vibration
- Human subject testing & surveys
 - bias, expensive, liability concerns



Human pressure map



Manikin pressure map

Ultimate goal:

Eliminate the need for human subjects in quantifying long term comfort by measuring and/or estimating the following parameters: pressure and shear force distributions, and seat vibration. Provide results that match human distribution of weight as closely as possible.

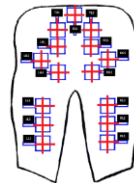
Device Design

Tissue Mimicking Component:

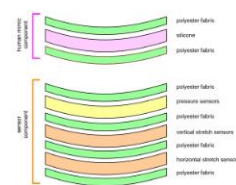
- Consists of a flat layer of Smooth-On Silicone with a 1A:1B mix ratio.

Sensor Suit Component:

- Pressure Data: Collected by fifteen Ohmite pressure sensors.
- Stretch Data: Collected by thirty rubber stretch cord sensors.
 - Two sensors were used in conjunction with a single pressure sensor.
 - Stretch sensors are oriented perpendicular to one another with the midpoint in the center of the pressure sensor.
- Shear Estimation: pressure and stretch readings can be combined to give an estimation of shear force at a point.
- Noise Reduction: A layer of non-conductive fabric separates each sensor layer.



Sensor Node Array on Butt-form



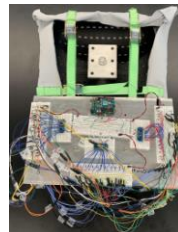
Layers of the device from innermost (top) to outermost (bottom)

Circuit Design:

- Code ran through Arduino IDE; circuitry wired to Arduino Uno board.
- Each sensor layer is connected to its own breadboard
- Inputs from voltage divider circuits are scanned with multiple analog channel multiplexers, then processed and displayed through Arduino.



Layers of the device: horizontal stretch sensors, vertical stretch sensors, pressure force sensors, silicone (left to right)



Finalized Full-Scale Prototype Device



Testing at Lear Comfortec Lab

Testing and Results

Silicone Properties

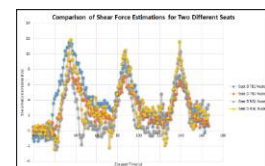
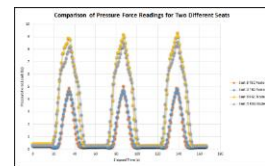
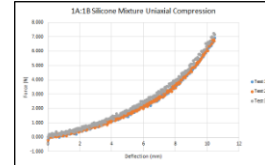
- Mechanical properties are similar to human tissue
 - Stiffness increases with higher load
- Human-like force distribution readings

Pressure Force Reading Results

- Compared 2 previously evaluated
 - Tailbone (TB2) and Right Glute (RG1) regions
- Higher pressure in gluteal region
- Slightly higher pressure in Seat B

Shear Estimation Results

- Compared 2 previously evaluated seats at different regions
 - Tailbone (TB2) and Right Glute (RG1) regions
- Similar across both regions and seats
- Very sensitive in stretch force readings



Conclusion and Future Work

- Pressure/stretch readings vary on different regions.
- Device is easily detachable and reusable for multiple seats.
- Graphical interface could visually show real-time results.
- Change the thickness/stiffness of the silicone layer to better mimic tissue.
- Use flat stretch sensors to prevent extra pressure on pressure sensor.
- Improve and streamline sensor calibration methods.
- Optimize shear calculation methods.
- Reduce noise and improve durability by using soldered breadboard.
- Minimize the gauge of the wiring while maintaining durability.

Acknowledgements

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